# Similar Triangles; Indirect Measurement

*Independent Set 6.07*

Name_____
Date_____
Period_____

*Six questions, on your own, in silence. Show the step that does the work.*

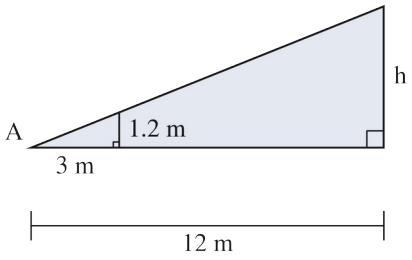
**1.** Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

**2.** Triangle PQR is similar to triangle XYZ. PQ is 8, QR is 10 and PR is 13, and XY is 20. Find YZ and XZ.

**3.** A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

**4.** A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

**5.** The two right triangles are similar and share corner A. The small one runs 3 m along the ground from A and stands 1.2 m high. The big one runs 12 m along the ground from A. How high is it?




**6.** Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.